

Harman Singh

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SUMMARY

Final-year engineering student graduating in May 2026, experienced in full-stack, backend, frontend and mobile development. Seeking 2026 new-grad software engineer roles.

EDUCATION

University of Guelph

Bachelor of Engineering in Engineering Systems and Computing

GPA: 3.8

Guelph, ON

Expected: May 2026

WORK EXPERIENCE

Citi

Software Engineer Intern

Mississauga, ON

May 2025 – August 2025

- Saved 5+ hours per week on compliance audits and enabled real-time dependency visibility by automating aggregation and tracking of dependencies across 90+ repositories with Python and Autosys.
- Automated real-time health alerts for services using Python, RabbitMQ, and MongoDB, eliminating the need for manual checks each night and reducing developer on-call disruptions.
- Reduced deploy time by up to 95% by implementing an OpenShift/Kubernetes Bash CLI to scale multiple services to the target pod count, consolidating 100+ redundant scripts and eliminating over 150 associated automated jobs.
- Created CI/CD pipelines to automate deploys using Tekton and Harness.

Ciena

Internal Tools Developer Intern

Ottawa, ON

May 2024 – Aug 2024

- Implemented a backend REST API using Python and FastAPI to streamline the process of managing infrastructure security vulnerabilities, resulting in a 30% reduction of vulnerabilities across company devices.
- Led project planning, scheduling, and backend/database development for an API used by the security team, coordinating with stakeholders to gather requirements and user input.
- Implemented authentication and authorization protocols using JWTs and Okta to safeguard backend resources.
- Automated builds and deploys using GitHub Actions, providing the security team with recent data on vulnerabilities.
- Deployed AWS API Gateways and Lambdas with Terraform to enable company-wide access to the application.

Wave Financial

Software Engineer Co-op

Toronto, ON

Jan 2024 – Apr 2024

- Developed and maintained accounting microservices, using Python and Django, improving system stability and UX.
- Optimized AWS S3 infrastructure by implementing lifecycle policies, resulting in \$2,000 in yearly cloud savings.
- Developed user interfaces for a mobile application with over 100,000 active users, utilizing React Native, HTML, CSS, JavaScript, TypeScript, and GraphQL to enhance user experience and improve application performance.
- Facilitated data-driven decision-making using Redshift and SQL to analyze 10,000+ data points for the ML team.
- Reduced incident response times by implementing automated Slack alerts using Datadog and Terraform.

Centre for Biodiversity Genomics

Programmer Co-op

Guelph, ON

Jun 2023 – Sep 2023

- Improved performance, enhanced security, and increased stability for the backend of a research institution with the world's largest repository of DNA barcode data (10M+ specimen records) by migrating from PHP 5.3 to PHP 8.1.
- Led the code review of the database access layer, resolving bugs and improving code maintainability.
- Refactored legacy PostgreSQL queries to enhance security and maintainability.

PROJECTS

Safe Walking App (polaris.harman-singh.com) | Python, FastAPI, React.js, AWS, Cloudflare

- Built a web app that finds safer walking routes in Toronto using aggregated crime data.
- Developed a REST API with Python/FastAPI and routing algorithms. Used caching to reduce response times by 90%.
- Designed and implemented a mobile-friendly UI with animations using React/Next.js and Tailwind CSS.
- Deployed on AWS EC2 with Nginx to manage traffic, protected via Cloudflare and rate-limiting to ensure low latency.

Campus Free Room Locator (gryphspace.com) | Python, React.js, Next.js, Cloudflare

- Developed a website to reduce crowding in study areas at UofG by locating unused classrooms to study in.
- Implemented using Python, Next.js/React.js frontend and hosted on Cloudflare.

SKILLS

Languages & Frameworks: Python, Django, FastAPI, HTML, CSS, React Native, React, JavaScript, TypeScript, SQL, Kubernetes, RabbitMQ, MongoDB, Nginx, GraphQL, C, C++, Java, Terraform, PHP, R, MATLAB

Tools & Concepts: AWS, REST APIs, Monitoring & Observability, Authentication, Object-oriented Programming, Fullstack, Frontend, Backend, Git, JIRA, Agile/Scrum, Datadog